Le réseau de transport d'électricité

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Towards unsupervised assessment of the accuracy of deep learning-based distributed PV mapping with open-source data

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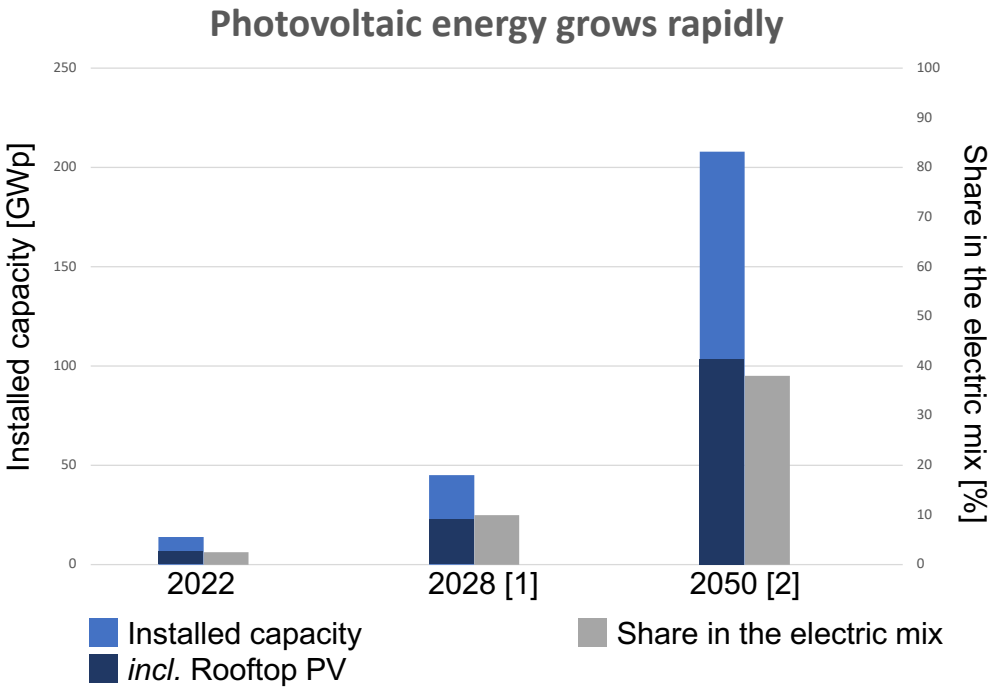
1 Deep learning to alleviate the missing knowledge on distributed solar installations

A growing « invisible » photovoltaic (PV) installed capacity



[1] *Programmation annuelle de l'énergie* (PPE)
[2] *Energy Pathways 2050* (2022), RTE France

PV energy is key in the energy transition.

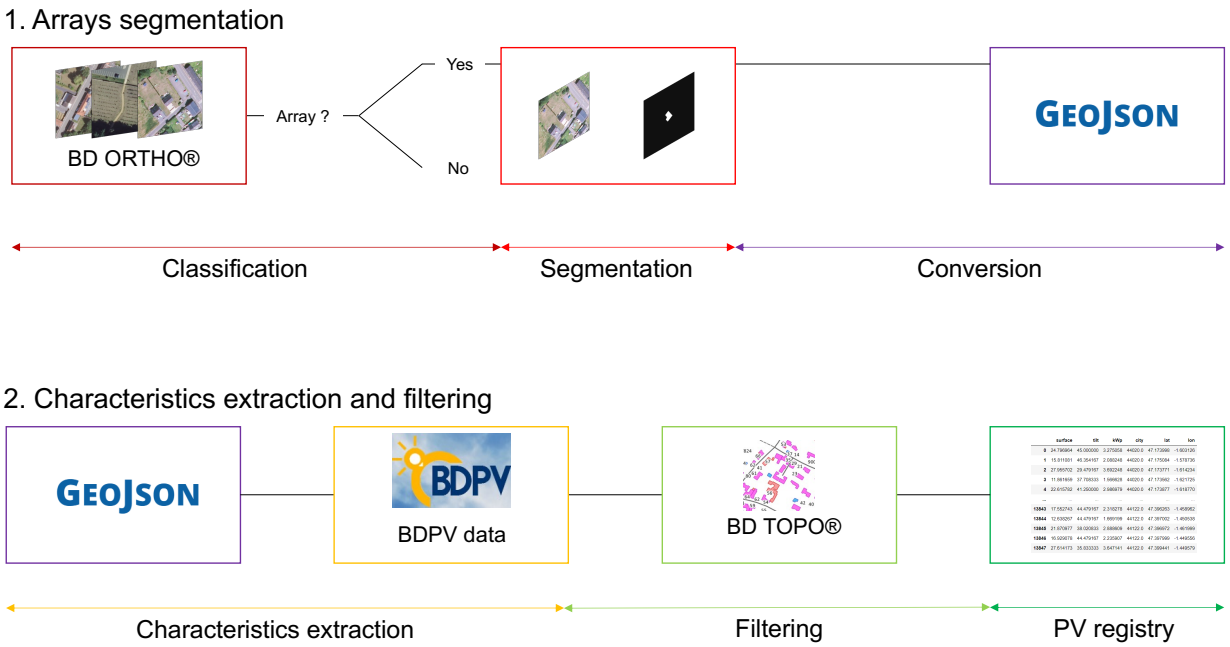
Rooftop PV: « **invisible » PV** (Shaker et al 2015).
➔ No measurements;
➔ No **reliable** characteristics database.

Unreliable estimates:
➔ Increased **congestions**, **overgeneration**, **imbalances** (Pierro et al 2022).

Need to map and characterize rooftop (distributed) PV installations

Leveraging deep-learning and EO data to map distributed PV

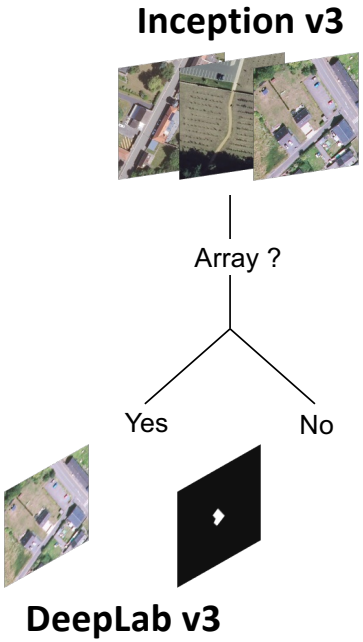
- We **build on existing works** (Yu et al, 2018; Mayer et al, 2022) to **map PV installations** from **overhead imagery**:



DeepPVMapper processing pipeline

DeepPVMapper: a two-step pipeline for automated PV mapping

1. Classification and segmentation

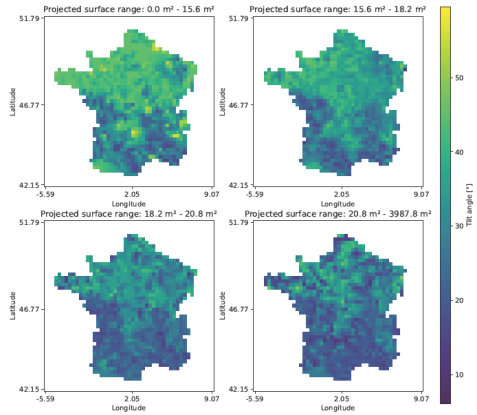


- Model backbones obtained from Mayer et al (2022);
- Fine tuning on our **new open source dataset** « BDAPPV » (Kasmi et al, 2022);

➢ Step 1 returns a set of **PV polygons**.

2. Characteristics extraction

- Surface:
  - ✓ Direct inference
- Tilt angle:
  - ✓ Look-up table
- Installed capacity:
  - ✓ Linear regression
- … and soon azimuth.



- **Reference data**: BDAPPV.

➢ Step 2 returns the **distributed PV database**.

Competitive accuracy compared to the state-of-the-art

1. Classification and segmentation

	Classification [F1]	Segmentation [IoU]	GSD [cm/px]
Mayer et al	0,87	0,74	10
Malof et al	-	0,67	30
Zech et al	0,82	-	10
Parhar et al	**0,97**	0,86	10
Ours	0,84	**0,86**	20

Classification and segmentation accuracy

2. Characteristics extraction

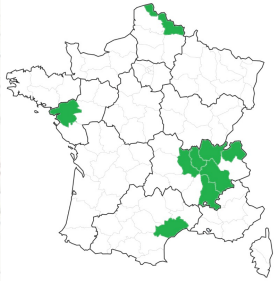
	MAE [kWp]	MAPE [%]
Mayer et al	1.92 – 24.84	27 - 37
Ours	**0.38**	**11,59**

Accuracy of the estimation of the installed capacity

Large scale mapping of PV installations over France

- **Largest automated distributed PV registry** with this level of detail to date.
  - *9 départements mapped in the north, south, west and south-east of France*

	surface	tilt	kWp	city	lat	lon
68738	11.895389	42.078104	1.889623	59084.0	50.683636	2.366848
26848	35.313887	30.970543	3.512614	1372.0	45.850301	5.671195
9362	35.806601	29.885555	3.184247	73129.0	45.597648	6.238806
80868	25.076319	37.598192	3.983459	59616.0	50.469836	3.556486
2988	15.959586	31.906815	2.574756	74282.0	46.022121	6.248841
1588	17.814072	31.515534	2.682169	74096.0	46.025674	6.102866
27736	29.512660	29.057030	3.226000	1160.0	46.258315	6.110942
84856	14.356873	20.634396	2.280638	34096.0	43.551525	3.169901
59668	17.172438	40.612170	2.727901	44006.0	47.445059	-2.450608
40756	12.879911	24.047517	2.046018	26088.0	45.029126	5.072853



Mapping area

Background: visualization of the city-wise installed capacity inferred from the raw registry.

How can we assess that these estimations are accurate?

2 *Downtream task accuracy*: evaluation of the outputs over the whole mapping area

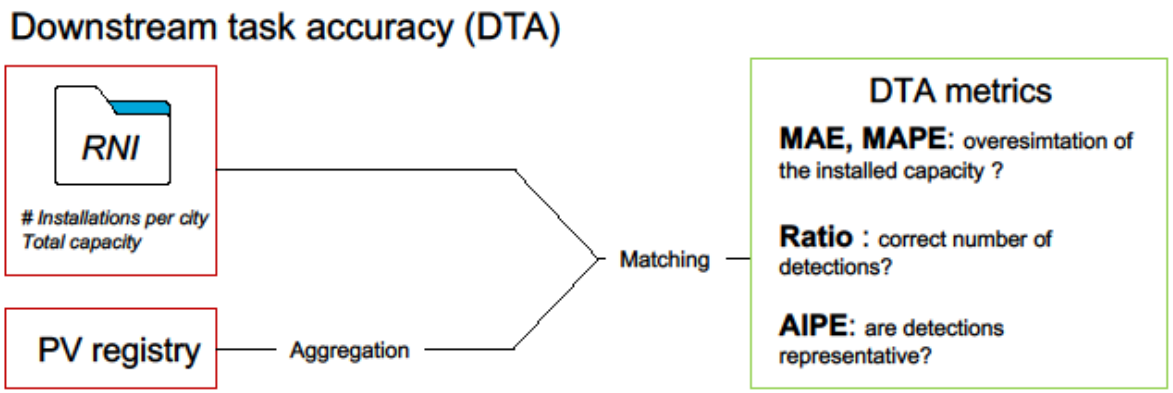
How can we evaluate automated PV registries' data?

- The **mapping area** is **orders of magnitude larger** than the training dataset.
- **Distribution shifts** issues can arise, **accuracy on the test set** may **not** be **representative** of the accuracy on the mapping area (Koh et al, 2020).
- We need to **evaluate the outputs' accuracy over the mapping area** (De Jong et al, 2020; Wiles et al, 2021).

BDAPPV training dataset (60 km²) **10 times**

Our proposal: the downstream task accuracy (DTA)

- Existing evaluation methods (Malof et al, 2019; Kausika et al, 2021) **require annotators** to validate the outputs. However, this is not satisfying as it is **not standardized** and **not representative**.
- The *Registre national d'installations* (RNI) compiles **city-wise aggregates** of distributed PV installations
  - ➢ Not detailed enough for PV mapping;
  - ➢ Detailed enough for **indirect accuracy evaluation**!



DTA

✓Uses the most exhaustive information available;

✓**Unsupervised**;

✓**Available over** (nearly) **all the mapping area**;

✓Expressed in end-user units.

Quantifying the performance gap

Baseline (DTA on the test set)

Dép.	MAPE [%]	med. APE [%]	mean Δ [-]	mean AIPE [%]	k_i [-]	$\hat{k}_i$ [-]	C_i [kWp]	$\hat{C}_i$ [kWp]
Test	17.61	-	0.92	-0.10	1485	1362	6473.8	5334,02

Best and worst « real » cases

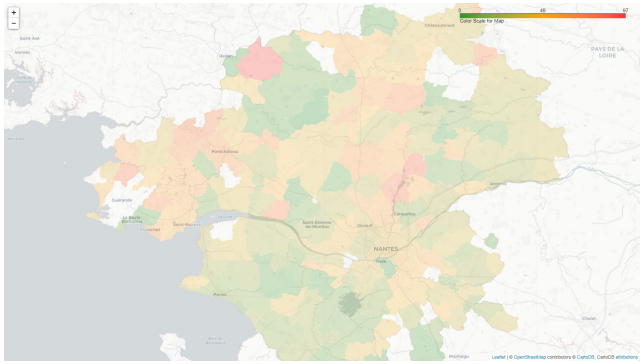
34	26.80 *(45.57)*	17.78 *(30.05)*	1.01 *(1.33)*	6.82 *(4.29)*	9199	8408 *(11445)*	35398.41	39897.16 *(52841.79)*
59	130.06 *(224.22)*	88.13 *(168.29)*	2.23 *(3.22)*	61.16 *(41.56)*	6453	9524 *(15393)*	22083.8	52790.18 *(73697.38)*

Average over the mapping area

Overall	47.45 *(66.20)*	32.81 *(30.66)*	1.03 *(1.46)*	16.33 *(12.03)*	72595	58818 *(84015)*	293120.70	280807.91 *(382967.83)*

30 percentage point performance gap!

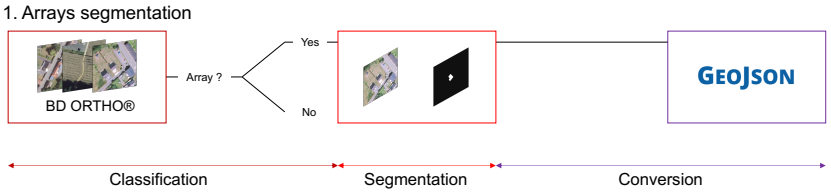
- DTA **unveils how a model truly performs** outside the training data.
- It enables end-users to **assess the quality of the data**, quantitatively and qualitatively.



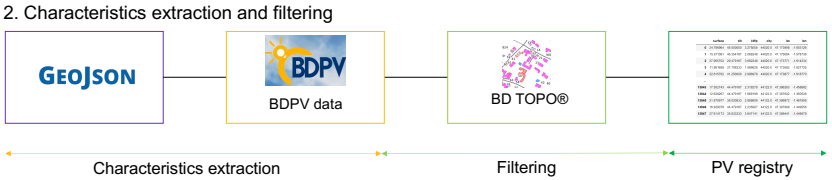
City-wise APE for the estimation of the installed capacity

Take-home messages

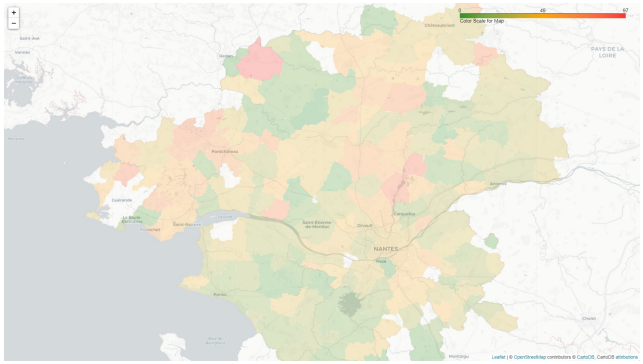
- We developped **DeepPVMapper**:
  - ✓ Automatically collects PV installations' characteristics;
  - ✓ We mapped more than 50000 km² over France.



- We introduce the **Downstream Task Accuracy** (DTA) to **assess the *true* accuracy of the outputs**
  - ✓ Indirect and unsupervised assessment of the accuracy on the mapping area;
  - ✓ The performance gap is about 30 percentage points.



- Future work:
  - Leverage the DTA to benchmark more advanced vision models (e.g., ViTs);
  - Expand DTA metrics to other characteristics;
  - Pursue the model's deployment, in France and beyond.



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Thank you!

Questions?

Project repository:

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Accuracy, cont'd

2. Characteristics extraction

	MRE [%]
Yu et al (2018)	2,1 – 3,0
Ours	**0.44**

Accuracy of the estimation of the surface

	ME [°]	MAE [°]
Ours	-2.40252	7.687603

Accuracy of the estimation of the tilt angle

All comparisons made on the test set, using the installation's metadata recorded in BDAPPV as a baseline.

MRE: mean relative error. See Yu et al (2018) for a definition.
ME/MAE : mean error, mean absolute error